

March 12, 1968

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3,372,499

PRESSING IRONS

Filed Dec. 15, 1966

FIG. 1

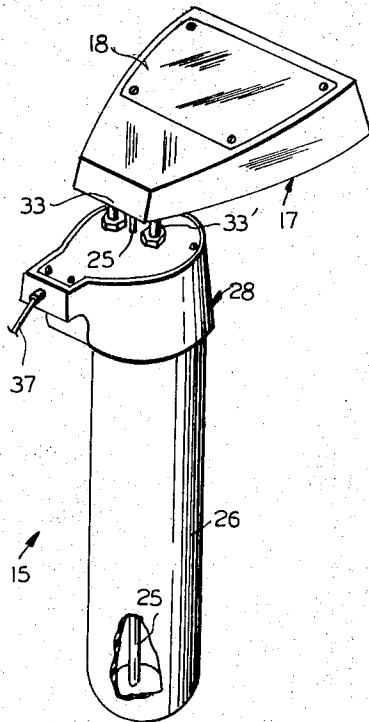


FIG. 2

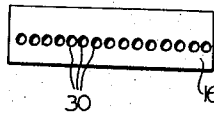


FIG. 3

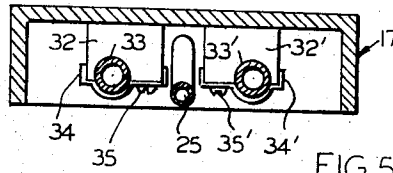
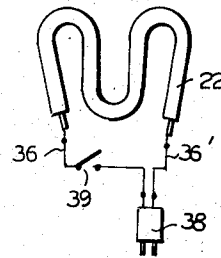
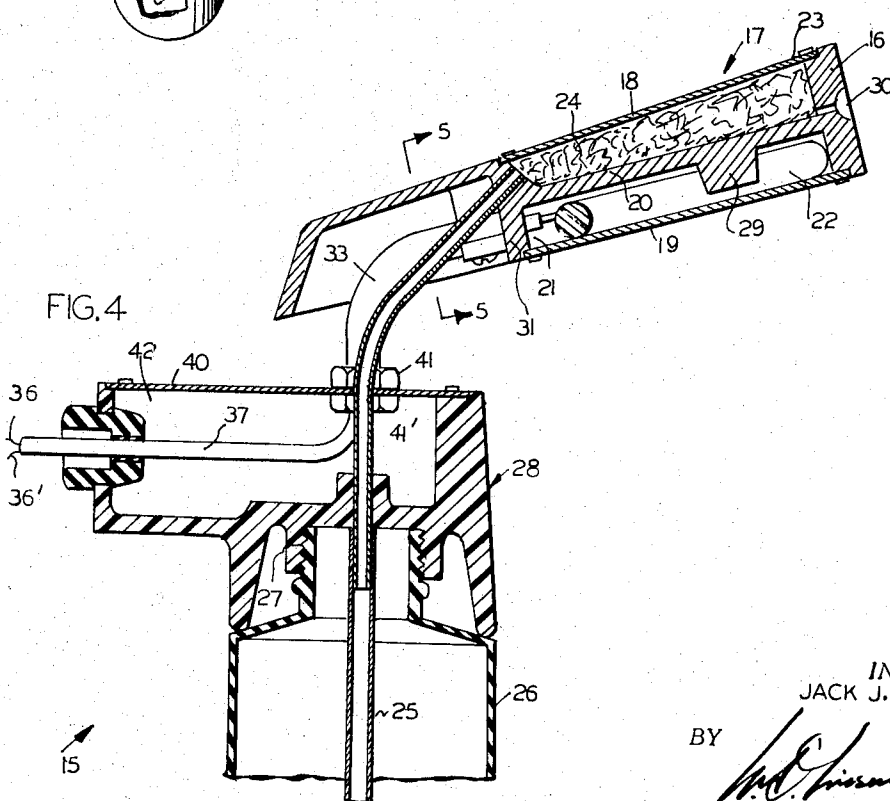


FIG. 5

FIG. 4



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3,372,499

PRESSING IRONS

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Filed Dec. 15, 1966, Ser. No. 601,928

9 Claims. (Cl. 38-77)

The present invention relates to, and its principal object is to provide a novel and improved hand-held, electrically-heated, steam-emitting pressing iron, adapted especially for touch-up work on a hanging garment. For example, this iron is to flatten wrinkles or to accentuate a crease in garments taken out of luggage or out of a closet or off a display rack. It is also useful to freshen trouser creases and for last minute work to be done on gowns being fitted.

Another object thereof is to provide a novel and improved pressing device of this type, in which steam generation is continuous though water is brought to the heating means in intermittent squirts.

A further object thereof is to provide a novel and improved electric steam pressing iron of the character mentioned, which is of compact and simple structure, reasonable in cost to manufacture, easy to use, and efficient in carrying out the purposes for which it is designed.

Other objects and advantages will become apparent as this disclosure proceeds.

For one practice of this invention, the pressing surface is offered by the perforated front wall of a comparatively thin casing which overhangs in a forwardly direction and is carried by the releasable head which is a screw-on cap for an upright water tank serving as the handle of the device. This tank is a resilient "squeeze" bottle having a threaded neck to receive said cap. Working said bottle by pinch and release movements of the handle-holding hand, causes water therein to be forced upwardly through a tube which discharges into a glass wool mass housed in said upper casing wherein an electrically-heated boiler vaporizes the water cached in the glass wool whereupon the steam generated is emitted from the perforations in the forward ironing surface.

In the accompanying drawing forming part of this specification, similar characters of reference indicate corresponding parts in all the views.

FIG. 1 is a perspective view of a pressing iron embodying the teachings of this invention. Part of the tank is shown broken through to expose the water delivery tube therein.

FIG. 2 shows the ironing surface.

FIG. 3 is a diagrammatic view of the electric heater element and the electrical circuit for its operation and control.

FIG. 4 is an enlarged central longitudinal section of the pressing iron shown in FIG. 1.

FIG. 5 is a section taken at lines 5-5 in FIG. 4.

The ironing device shown in the drawing as an example of this invention, is designated generally by the numeral 15. The ironing surface is the outer face of the flat front wall 16 of a metal casing denoted generally by the numeral 17. This casing has flat top and bottom walls indicated respectively as 18 and 19, between which there is a metal partition 20, dividing the casing into a lower chamber 21 for housing the electric heater element 22, and an upper compartment 23 filled with glass wool 24, in which water forced up the delivery tube 25 from the tank 26, is discharged. Said tank, serving as the handle for the device 15, is an upright "squeeze" bottle preferably of polyethylene, that is, said bottle is deformable by pinching it, but due to its resiliency, will upon release, reassume its original shape. The neck of this bottle is exteriorly threaded as shown at 27, to receive the head indicated generally as 28, as a screw-on cap. The water delivery tube

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25, extends from near the bottom of the tank 26, up through the head 28, and thence to the space 23, so water sent into the glass wool mass 24, will be vaporized by the hot casing and particularly by the hot plate, the partition 20. The steam thus generated is discharged through the perforations 30, in the casing's front wall 16; said perforations being communicative with the space 23. The glass wool mass 24 is preferably of sufficient quantity to hold a supply of water in excess of the rate of evaporation, so the steam generated is a constant flow, and not in pulsations. The mass 24 may be any fibrous material which preferably is rust-proof.

Downwardly from the partition 20, is a central boss 29 which fits into the center U-form of the heating element 22, to locate it within the space 21. Rearwardly from the rear wall 31 of the casing 17, are two spaced saddle bosses 32, 32' to seat the elbows 33, 33' which are mounted thereon, by for instance, the straps 34, 34' held in place by the screws 35, 35' respectively. Through these elbows extend the electrical conductors 36, 36' which are connected to the terminals of the heating element 22; said conductors being in a cord 37, terminating in a connection plug 38. In one of said conductors, a switch 39 may be interposed to control the circuit. Said elbows, exteriorly threaded at their lower ends, extend through suitable holes in a plate 40, whereat they are secured by the nuts 41, 41' as shown in FIG. 4, and thus mount the casing onto the head 28. The upper end of the delivery tube 25 is tight within an opening through the casing's rear wall 31, and discharges into the upper compartment of the casing filled with the glass wool 24. The discharge end of said tube 25 may be flattened or otherwise provided to cause a spray discharge, which is readily understood without further illustration. The casing 17 is preferably tilted forwardly upwardly a bit as shown, and its ironing wall 16 is preferably smooth and polished. The head 28 may be formed with a connection chamber 42, in which the switch 39 can be housed with its operating element exposed to be accessible, and though not specifically shown, is believed readily understandable without the need of further illustration.

The top and bottom walls 18, 19 of the casing 17, and the cover plate 40, may be of sheet metal suitably held in place by screws. The remainder of the body of said casing 17, may be of cast aluminum or other metal. The body of the head 28 may be of Bakelite, a ceramic or other suitable heat-insulative material. The tank 26 may be of rubber, or polyethylene as mentioned.

For practical dimensions, the heating surface 16 may measure one inch in height and four inches in length, and the height of the tank 26 may be about eight inches.

To use the device, the garment to be worked on is hung up on a hanger high upon a nail extending from the front surface of a door, or depending from any other suitable raised support. Below the region to be pressed, the garment is pulled taut. The iron of course has been made hot by closing the switch 39 in a powered circuit, and the tank 26 has been nearly filled with water. Using the tank as a handle, the device is held in hand and the ironing surface 16 is applied to the garment while the user manipulates his fingers for a grasp and release action. This will on each squeeze, cause water to be forced up the delivery tube 25, and discharged into the glass wool 24, wherefrom it is converted into steam which is ejected through the openings 30 onto the garment being pressed. If no steam is desired, the tank 26 is not squeezed. The underside 19 of the casing 17, may be used for ironing on the shoulder of the garment, and so may the surface 16 be used for such purpose.

A special use for this ironing device, is for touching up garments being worn and fitted at the last minute for bridal occasions, in dressing rooms.

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If desired, a small wood block held in hand, may be used as an ironing board.

Upon every squeeze of the tank 26, there will be a squirt of water into the chamber 23. If the rate of vaporization equals or is of a capability to exceed the water fed to the glass wool 24, then the steam emitted from the openings 30 will be in pulsations. However, if before the heater 22 is actuated, the tank is worked to fill the chamber 23, to saturate the glass wool 24, and is continuously worked thereafter, a boiler having a water supply is created, so the emission of steam will be in a constant flow through the openings 30.

This invention is capable of numerous forms and various applications without departing from the essential features herein disclosed. It is therefore desired and intended that the embodiment shown herein shall be deemed merely illustrative and not restrictive, and that the patent shall cover all patentable novelty herein set forth; reference being had to the following claims rather than to the specific description and showing herein, to indicate the scope of this invention.

I claim:

1. In an ironing device of the character described, a casing having a metal front wall whose outer surface serves as the ironing surface, an electric heater element behind said front wall and carried by the casing, a resilient squeeze bottle behind said front wall, serving as a water tank and as the handle of said device, a cap closing said bottle and carrying said casing, a water-delivery tube extending from within the lower part of said bottle, then outwardly thereof in water-tight relation therewith, and terminating within said casing wherein it is adapted to discharge, whereupon squeezing said bottle, water therefrom will be forced through said tube and into the casing; the heat generated by the heater element when actuated, being for heating said casing and for vaporizing the water entered into said casing; said front wall having perforations through which the steam made will pass out of the casing.

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2. A device as defined in claim 1, including a water-absorbing mass such as glass wool, within said casing to receive the water discharged from said tube.

3. A device as defined in claim 2, wherein said mass is of such quantity that it is capable of holding more water than the rate of vaporization of the heating element so the flow of steam is continuous while water is delivered to said casing in squirts.

4. A device as defined in claim 2, including a metal partition wall within the casing forming upper and lower compartments therein; the heater element being in one of said compartments, and the delivery tube discharging into the other compartment; said mass being in such other compartment.

5. A device as defined in claim 1, wherein the bottle is an elongated upright receptacle extending lower than the casing, and the ironing surface is substantially upright.

6. A device as defined in claim 1, wherein the front wall of the casing is rectangular.

7. A device as defined in claim 1, wherein the casing has a flat bottom wall of metal, whose bottom surface is adapted to serve as an ironing surface.

8. A device as defined in claim 1, wherein the cap is releasably mounted on the bottle, and the water-delivery tube is through the cap.

9. A device as defined in claim 1, wherein the bottle is upright and the ironing surface of the casing is flat and has a slight slant forwardly upwardly.

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